

● HARD

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

06

10 answers · Rationale-rich · Teach from doc

Q1**D**

D. The time at which the projectile hits the ground

Choice D is correct. The positive x-intercept of the graph of $y = h(x)$ is a point (x, y) for which $y = 0$. Since $y = h(x)$ models the height above the ground, in feet, of the projectile, a y-value of 0 must correspond to the height of the projectile when it is 0 feet above ground or, in other words, when the projectile is on the ground. Since x represents the time since the projectile was launched, it follows that the positive x-intercept, $(x, 0)$, represents the time at which the projectile hits the ground.

Choice A is incorrect and may result from misidentifying the y-intercept as a positive x-intercept. Choice B is incorrect and may result from misidentifying the y-value of the vertex of the graph of the function as an x-intercept. Choice C is incorrect and may result from misidentifying the x-value of the vertex of the graph of the function as an x-intercept.

Q2**336**

The correct answer is 336. By the zero product property, if $x + 14t - x = 0$, then $x + 14 = 0$, which gives $x = -14$, or $t - x = 0$, which gives $x = t$. Therefore, $gx = 0$ when $x = -14$ and when $x = t$. Since the graph of $y = gx$ passes through the point $(24, 0)$, it follows that $g(24) = 0$, so $t = 24$. Substituting 24 for t in the equation $gx = x + 14t - x$ yields $gx = x + 14(24) - x$. The value of $g(0)$ can be calculated by substituting 0 for x in this equation, which yields $g(0) = 0 + 14(24) - 0$, or $g(0) = 336$.

Q3**A****A.** -13

Choice A is correct. It's given that $gx = fx + 5$. Since $fx = 4x^2 + 64x + 262$, it follows that $fx + 5 = 4x^2 + 64x + 5 + 262$. Expanding the quantity $x + 5^2$ in this equation yields $fx + 5 = 4x^2 + 10x + 25 + 64x + 5 + 262$. Distributing the 4 and the 64 yields $fx + 5 = 4x^2 + 40x + 100 + 64x + 320 + 262$. Combining like terms yields $fx + 5 = 4x^2 + 104x + 682$. Therefore, $gx = 4x^2 + 104x + 682$. For a quadratic function defined by an equation of the form $gx = ax - h^2 + k$, where a , h , and k are constants and a is positive, gx reaches its minimum, k , when the value of x is h . The equation $gx = 4x^2 + 104x + 682$ can be rewritten in this form by completing the square. This equation is equivalent to $gx = 4x^2 + 26 + 682$, or $gx = 4x^2 + 26x + 169 - 169 + 682$. This equation can be rewritten as $gx = 4x + 13^2 - 169 + 682$, or $gx = 4x + 13^2 - 4169 + 682$, which is equivalent to $gx = 4x + 13^2 + 6$. This equation is in the form $gx = ax - h^2 + k$, where $a = 4$, $h = -13$, and $k = 6$. Therefore, gx reaches its minimum when the value of x is -13.

Choice B is incorrect. This is the value of x for which fx , rather than gx , reaches its minimum.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $fx - 5$, rather than $fx + 5$, reaches its minimum.

Q4**C****C.** $fx = x^2 + 91x$

Choice C is correct. It's given that the function f gives the product of a number, x , and a number that is 91 more than x . A number that is 91 more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**A****A.** $(0, 14)$

Choice A is correct. The y -intercept of the graph of $y = fx$ in the xy -plane occurs at the point on the graph where $x = 0$. In other words, when $x = 0$, the corresponding value of fx is the y -coordinate of the y -intercept. Substituting 0 for x in the given equation yields $f0 = -82^0 + 22$, which is equivalent to $f0 = -81 + 22$, or $f0 = 14$. Thus, when $x = 0$, the corresponding value of fx is 14. Therefore, the y -intercept of the graph of $y = fx$ in the xy -plane is 0, 14.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This could be the y -intercept for $fx = -82^x$, not $fx = -82^x + 22$.

Q6**3.44**

Accept also: 86/25

The correct answer is 3.44. It's given that $f0 = 86$ and that for each increase in x by 1, the value of fx decreases by 80%. Because the output of the function decreases by a constant percentage for each 1-unit increase in the value of x , this relationship can be represented by an exponential function of the form $fx = ab^x$, where a represents the initial value of the function and b represents the rate of decay,

expressed as a decimal. Because $f0 = 86$, the value of a must be 86. Because the value of fx decreases by 80% for each 1-unit increase in x , the value of b must be $(1 - 0.80)$, or 0.2.

Therefore, the function f can be defined by $fx = 860.2^x$. Substituting 2 for x in this function yields $f2 = 860.2^2$, which is equivalent to $f2 = 860.04$, or $f2 = 3.44$. Either 3.44 or 86 / 25 may be entered as the correct answer.

Alternate approach: It's given that $f0 = 86$ and that for each increase in x by 1, the value of fx decreases by 80%. Therefore, when $x = 1$, the value of fx is $(100 - 80)\%$, or 20%, of 86, which can be expressed as 0.2086. Since $0.2086 = 17.2$, the value of $f1$ is 17.2. Similarly, when $x = 2$, the value of fx is 20% of 17.2, which can be expressed as 0.2017.2. Since $0.2017.2 = 3.44$, the value of $f2$ is 3.44. Either 3.44 or 86 / 25 may be entered as the correct answer.

Q7**C****C. 970**

Choice C is correct. It's given that a quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. This quadratic function can be defined by an equation of the form $f(x) = ax - h^2 + k$, where $f(x)$ is the height of the object x seconds after it was launched, and a , h , and k are constants such that the function reaches its maximum value, k , when $x = h$. Since the model indicates the object reaches its maximum height of 1,034 feet above the ground 8 seconds after being launched, $f(x)$ reaches its maximum value, 1,034, when $x = 8$. Therefore, $k = 1,034$ and $h = 8$. Substituting 8 for h and 1,034 for k in the function $f(x) = ax - h^2 + k$ yields $f(x) = ax - 8^2 + 1,034$. Since the model indicates the object has an initial height of 10 feet above the ground, the value of $f(x)$ is 10 when $x = 0$. Substituting 0 for x and 10 for $f(x)$ in the equation $f(x) = ax - 8^2 + 1,034$ yields $10 = a(0) - 8^2 + 1,034$, or $10 = 64a + 1,034$. Subtracting 1,034 from both sides of this equation yields $64a = -1,024$. Dividing both sides of this equation by 64 yields $a = -16$. Therefore, the model can be represented by the equation $f(x) = -16x - 8^2 + 1,034$. Substituting 10 for x in this equation yields $f(10) = -16(10) - 8^2 + 1,034$, or $f(10) = 970$. Therefore, based on the model, 10 seconds after being launched, the height of the object above the ground is 970 feet.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**B****B.** 45

Choice B is correct. It's given that $fx = x^2 - 44x - 46$, which can be rewritten as $fx = x^2 - 90x + 2,024$. Since the coefficient of the x^2 -term is positive, the graph of $y = fx$ in the xy -plane opens upward and reaches its minimum value at its vertex. For an equation in the form $fx = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $fx = x^2 - 90x + 2,024$, $a = 1$, $b = -90$, and $c = 2,024$. It follows that the x -coordinate of the vertex is $-\frac{-90}{2(1)}$, or 45. Therefore, fx reaches its minimum when the value of x is 45.

Choice A is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = fx$ in the xy -plane.

Choice C is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = fx$ in the xy -plane.

Choice D is incorrect. This is the y -coordinate of the vertex of the graph of $y = fx$ in the xy -plane.

Q9**D****D.** 8

Choice D is correct. It's given that the graph of $y = h(x)$ intersects the x -axis at $(0,0)$ and $(t,0)$, where t is a constant. Since this graph intersects the x -axis when $y = 0$ or when $h(x) = 0$, it follows that $h(0) = 0$ and $h(t) = 0$. If $h(t) = 0$, then $0 = 2(t-4)^2 - 32$. Adding 32 to both sides of this equation yields $32 = 2(t-4)^2$. Dividing both sides of this equation by 2 yields $16 = (t-4)^2$. Taking the square root of both sides of this equation yields $4 = t-4$. Adding 4 to both sides of this equation yields $8 = t$. Therefore, the value of t is 8.

Choices A, B, and C are incorrect and may result from calculation errors.

Q10

D

D. -12

Choice D is correct. The equation of a parabola in the xy -plane can be written in the form $y = ax^2 - h^2 + k$, where a is a constant and h, k is the vertex of the parabola. If a is positive, the parabola will open upward, and if a is negative, the parabola will open downward. It's given that the parabola has vertex $9, -14$. Substituting 9 for h and -14 for k in the equation $y = ax^2 - h^2 + k$ gives $y = ax^2 - 9^2 - 14$, which can be rewritten as $y = ax^2 - 18x + 81 - 14$, or $y = ax^2 - 18ax + 81a - 14$. Distributing the factor of a on the right-hand side of this equation yields $y = ax^2 - 18ax + 81a - 14$. Therefore, the equation of the parabola, $y = ax^2 - 18ax + 81a - 14$, can be written in the form $y = ax^2 + bx + c$, where $a = a$, $b = -18a$, and $c = 81a - 14$. Substituting $-18a$ for b and $81a - 14$ for c in the expression $a + b + c$ yields $a + -18a + 81a - 14$, or $64a - 14$. Since the vertex of the parabola, $9, -14$, is below the x -axis, and it's given that the parabola intersects the x -axis at two points, the parabola must open upward. Therefore, the constant a must have a positive value. Setting the expression $64a - 14$ equal to the value in choice D yields $64a - 14 = -12$. Adding 14 to both sides of this equation yields $64a = 2$. Dividing both sides of this equation by 64 yields $a = \frac{2}{64}$, which is a positive value. Therefore, if the equation of the parabola is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants, the value of $a + b + c$ could be -12 .

Choice A is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants and $a + b + c = -23$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the x -axis at zero points, not two points.

Choice B is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants and $a + b + c = -19$, then the value of a will be negative, which means the parabola will open downward, not upward, and will intersect the x -axis at zero points, not two points.

Choice C is incorrect. If the equation of a parabola with a vertex at $9, -14$ is written in the form $y = ax^2 + bx + c$, where a, b , and c are constants and $a + b + c = -14$, then the value of a will be 0 , which is inconsistent with the equation of a parabola.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

